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Research papers

Operational dynamics of packed-bed thermal energy storage: A novel approach to monitor its thermal state☆

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ABSTRACT

Heating solids arranged in a packed bed is a simple method to store sensible thermal energy. This study focuses on cylindrical packed-bed configurations, where a temperature profile develops along the stacked solids and moves towards the system's outlet. Therefore, the availability to store thermal energy in the packed bed decreases over time and the exhaust energy outside the storage increases. To address this challenge, the present work introduces a novel operational metric to monitor the thermal state of packed-bed storage systems. This operational metric relates the stored potential in the system with an ideal stage, and its definition enables constraining the system's operation with a simple mathematical condition, which is a significant outcome from the traditional approach of using arbitrary temperature limits to stop the charging process. Its calculation requires the measurement of the top and bottom temperatures of the packed bed and knowing the system's maximum operating temperatures. When applying the mathematical condition, the required charging time to reach that stage and the cut-off temperature can be obtained. A parametric analysis correlated these operating quantities to design data to predict their value given the design conditions. This work provides a new perspective on the dynamic operation of cylindrical packed-bed sensible thermal energy storage systems, offering a simple yet effective strategy to enhance system performance.

1. Introduction

Packed-bed thermal energy storage systems (PBTES) have been proposed as a promising technology in a wide range of operating temperatures and applications [1,2]. This storage method has also demonstrated a potential reduction of thermal energy storage (TES) implementation costs compared to other commercial TES technologies [3]. PBTES uses particulate solids as the storage medium located inside a container tank. During the charging operation, the excess heat produced in a thermal source is transported to the storage through a heat transfer fluid (HTF). When the stored thermal energy is needed, the system is discharged by flowing low-temperature HTF from a heat sink to the PBTES, recovering the thermal energy contained and supplying it to the thermal load (see Fig. 1). Inside the storage tank, the fluid flows through the solids, conveying heat; as a result, a temperature profile, called thermocline, evolves along the flow direction, separating the storage medium's hot and cold zones. Consequently, the charge and discharge processes occur in different directions to avoid mixing from finite temperature differences between streams and are only affected by conductive losses [4,5]; therefore, the thermocline moves forward and backward within the tank as those processes continue [6].

Due to the temperature profile developed inside the PBTES, thermal management of its operation is crucial to guarantee a performance that fits the needs of the thermal demand. This is, in general, a desired discharge temperature during a design time. In that regard, introducing operational constraints into the sizing procedure of PBTES systems has become relevant for most sensitivity analyses and optimization studies available in the literature [7,8]. These constraints are mainly associated with the PBTES outflow temperature, which performs a non-negligible dynamic during the PBTES operation, as shown in Fig. 2. In the charging process, the outflow temperature remains constant while the bed is heated; then, the outflow temperature experiences an exponential increase due to the thermocline release since all the solids have been heated above the reference temperature [6]. Conversely, the

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Nomenclature

c_p	Specific heat (J kg^{-1} K^{-1})
d_p	Particle diameter (m)
h	Specific enthalpy (J kg^{-1})
h_v	Volumetric heat transfer coefficient (W/m^{-3}/K)
k	Thermal conductivity (W m^{-1} K^{-1})
$\dot{m}$	Mass flow rate (kg s^{-1})
s	Specific entropy (J kg^{-1} K^{-1})
t	Time (s)
z	Axial coordinate (m)
A_{pb}	Cross-sectional area of the TES (m^2)
L	Length of the TES in the axial direction (m)
N_s	Entropy generation number (–)
Nu	Nusselt number (–)
Pr	Prandtl number (–)
$\dot{Q}_{load}$	Thermal demand/load (W)
Re	Reynolds number (–)
S_{gen}	Entropy generation (J)
T	Temperature (K)
V	Effective solid volume (m^3)
V_{TES}	TES volume (m^3)
X	Exergy (J)

Greek symbols

α	Correlation constant (s)
β	Correlation constant (s)
δ	Thermocline thickness (–)
η_0	Design storage efficiency (–)
η_{rt}	Round-trip efficiency (–)
μ	Dynamic viscosity (Pa s)
ρ	Density (kg/m^3)
τ	Time constant (s)
ε	Void fraction (–)
ζ	Utilization factor (–)

Subscripts

0	Reference state
B	Related to the bottom part of the PBTES
ch	Related to the charging process
co	Cut-off limit
dch	Related to the discharging process
des	Related to a design parameter
$dest$	Destruction loss term
eff	Effective value
eq	Related to equivalent properties
f	Related to the fluid phase
H	High-temperature source
L	Low-temperature sink
$loss$	Loss exergy/energy term
max	Related to a maximum value
min	Related to a minimum value
$outflow$	Related to the outflow temperature
s	Related to the solid phase
T	Related to the top part of the PBTES
$threshold$	Related to a threshold temperature value

Abbreviations

AR	Aspect Ratio
BC	Boundary Condition
CSP	Concentrating Solar Power
HTF	Heat Transfer Fluid
MTTES	Multi-tank Thermal Energy Storage
NLS	Non-linear Least Squares
NTU	Number of Heat Transfer Units
PBTES	Packed-bed Thermal Energy Storage
PLC	Programmable Logic Controller
SOC	State of Charge
STP	State of Thermal Potential
TCCM	Thermocline Control Methods
TES	Thermal Energy Storage

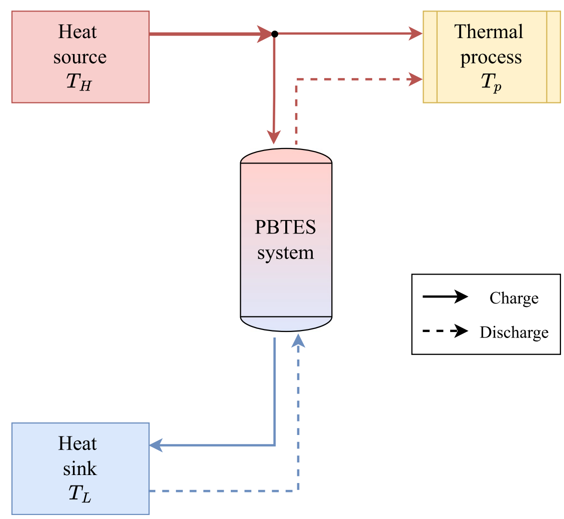

Fig. 1. Basic operation of a TES system.

discharging outflow temperature behavior is similar but with a decrease in the high temperatures released.

Different thermocline control methods (TCCM) have been investigated to provide a steady operation in the outflow temperature. According to Ref. [9], there are passive and active TCCMs depending on whether they are focused on the storage design or its operation. Moreover, Ref. [10] assessed some active TCCMs and compared the extraction, injection, and mixing methods in terms of their capability to develop a steeper thermocline and a constant outflow temperature. Furthermore, thermocline control would benefit the utilization factor ζ, which indicates the actual use of the storage capacity between charge and discharge periods [9,11]. However, the technical challenges of these methods are that their design considers vertical stacking of the packed elements, leading to tall storage and relevant structural requirements to support the weight of the stacked beds [12]. In that context, multi-tank TES (MTTES) can address those challenges if the packed-bed units are placed next to each other. As a result, Ref. [12] researched the MTTES based on a serial operating mode, using the mixing and extracting TCCM previously proposed by Ref. [10] for stacked PBTES. Through this configuration, the authors demonstrated the low drop in the outflow temperature during the discharging process, which is relevant for many thermal applications that require a stable operation. Optimization studies of this configuration are still needed to

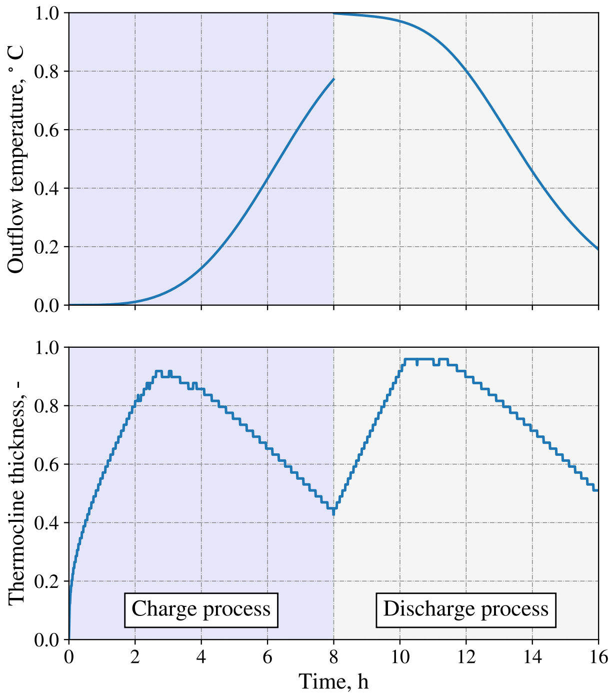

Fig. 2. Dynamics on the outflow temperature and thermocline thickness.

exploit its benefits by selecting the best set of target temperatures to maximize the system's utilization factor and reduce thermal losses.

Despite the vast development of design and operational configuration studies, there is a lack of metrics to monitor the operational behavior of PBTES systems. Measuring the outflow temperature can provide useful information, but without a target cut-off temperature, the variable itself does not provide end-of-charge and discharge criteria. Ref. [13] developed a comprehensive analysis of how the round-trip efficiency is affected by the charge and discharge cut-off temperature, maximizing its value when small cut-off temperatures at charge are selected. Likewise, Ref. [14] confirmed that result for different PBTES container designs, whereas the utilization factor can be maximized by increasing the charge–discharge depth. These studies do not consider the combined effect of varying the inlet mass flow rate and the desired cut-off temperatures on the PBTES performance, which could resemble a possible scenario for the integration of a PBTES into a Concentrating Solar Power (CSP) plant operation [8]. Overall, through these assessments, the operating times for the charge and discharge can be obtained; hence, a stop criterion for the PBTES operation can be defined.

This work aims to develop a monitoring metric that contributes to defining the thermal state of a PBTES system. This metric will provide information to manage the PBTES performance with different design schemes, i.e., single or multi-tank systems; therefore, with input design and operating characteristics, this metric can determine the target temperatures to end the charging process. In that context, the article contributes to the thermal management of the PBTES's dynamic behavior and performs an initial approach to optimum control characteristics. First, the article provides a general description of the domain and boundary conditions at Section 2. A brief literature review is included to discuss the different approaches to monitoring the thermal state of PBTES systems. Then, it is followed by the proposed method, which includes its mathematical definition and the information that it provides. The metrics are compared in Section 2.3 based on their time response and performance during a charge–discharge process using different stopping conditions for the PBTES operation. Section 3 presents the parametric analysis and the methods to correlate operational parameters with design inputs. That section includes definitions of the functions to calculate the operational parameters, and it discusses performance prediction by comparing the values obtained from the correlation and the ones obtained from the numerical simulation of the constrained operation. Finally, the article summarizes the potential applications of this monitoring approach and how it can be employed in existing test benches, numerical simulations, or optimization studies.

2. Mathematical modeling

The domain for the analysis consists of a packed bed of rocks arranged in a cylindrical container. This container has inlet ports at the top and bottom lids to enable the mass and energy transport from a heat transfer fluid (HTF). As observed in Fig. 1, the integration of the PBTES is located between two thermal reservoirs; hence, the HTF flows from the heat source or the heat sink, depending on the process developed. During the charging process, the arbitrary heat source provides thermal energy to the PBTES by supplying HTF at T_H. In time, the fluid exits the bottom part of the tank at $T_L < T_{outflow,\ ch} < T_H$ and can be exhausted to the heat sink. Conversely, in the discharging process, a flowing stream from the heat sink is injected into the PBTES to recover the stored thermal energy and supply it to the thermal process. The outflow temperature now is $T_{outflow,\ dch} > T_L$.

Due to the thermocline development, it is ideal to have a high-temperature difference between the top and bottom of the tank, with a maximum at $\Delta T = T_H - T_L$; therefore, the outflow temperatures at charge and discharge should be equal to those low- and high-temperature limits. However, the conduction heat transfer and thermal losses cause the thermocline to spread along the storage tank, which triggers its early extraction, i.e., $T_{outflow,\ ch} > T_L$ or $T_{outflow,\ dch} < T_H$. Keeping the thermocline inside the storage system allows a stable heat supply during the discharge and increases the system's utilization factor [9]. Consequently, having information about the energy content in the PBTES or the thermocline development is relevant for managing the system's operation since measuring these provides information about the PBTES's thermal state, which is crucial to managing the system's operation. The following subsection explores the different approaches to monitoring the thermal state of PBTES and the useful information they provide.

2.1. Current monitoring indicators to define the thermal state of PBTES

Outflow temperature. The simplest way to monitor the storage operation is to measure the outflow temperature. According to Ref. [9], limiting the outflow temperature during the charging process is relevant for CSP plant operation to reduce the components' wear and increase their lifetime. In addition, power block operating conditions constrain the outflow temperature during the discharge process [9]. In this context, defining a cut-off temperature difference (ΔT_{co}) as an operating parameter helps the implementation of control logic schemes [10, 13], i.e.,

$$\Delta T_{co,\ ch} = T_{outflowf,\ ch,\ max} - T_L \tag{1}$$

$$\Delta T_{co,\ dch} = T_H - T_{outflow,\ dch,\ min} \tag{2}$$

where the subscripts ch and dch refer to the charge and discharge processes, $T_{outflow,\ ch,\ max}$ corresponds to the maximum permissible temperature that the system's components can admit, and $T_{outflow,\ dch,\ min}$ is the minimum stream temperature that the demand requires.

The outflow temperature itself does not provide information about the useful energy contained in the storage tank; the constraints are only related to the PBTES boundary conditions. Therefore, maximizing the usage of the PBTES implies increasing the cut-off temperatures [14], but the thermocline is also discharged from the system.

Thermocline thickness. The measurement of thermocline thickness refers to monitoring its physical thickness within a specific temperature range [15]. The user defines the thresholds according to the operational parameters for the PBTES. In general, the thermocline thickness is calculated as

$$\delta = \frac{1}{L}\left(z\Big|_{T=T_L+T_{threshold,B}} - z\Big|_{T=T_H-T_{threshold,T}} \right) \tag{3}$$

where L is the PBTES length, $T_{threshold}$ is the limit temperature for the thermocline development located at the bottom and the top part of the tank (denoted by the subscripts B and T), and z the axial coordinate of the PBTES system. Fig. 2 shows how the thermocline thickness varies in time. In contrast with the outflow temperature, the thermocline thickness reaches a maximum at a specific time, indicating that the thermocline has reached the PBTES outlet and is being released from the storage tank. Thermocline thickness provides information on when to end a charge or discharge process based on its maximum value; therefore, it provides insights into how the system must be operated to minimize thermocline release. Nevertheless, measuring its value requires building the temperature profile accurately by using several temperature sensors. Another option is to perform a linear interpolation between temperature measurements, but it increases uncertainty [16].

State of charge. The state of charge (SOC) measures the TES's instantaneous energy content against its maximum design capacity [17], i.e.,

$$\begin{aligned} \mathrm{SOC}(t) = & \frac{\int_V (1-\varepsilon)\ \left(\rho c_p\right)_s\ \left(T_s(t,\ V) - T_0\right) dV}{\left(\rho c_p\right)_{eq}\ V_{TES}\ \left(T_H - T_0\right)} \\ & + \frac{\int_V \varepsilon\ \left(\rho c_p\right)_f\ \left(T_f(t,\ V) - T_0\right) dV}{\left(\rho c_p\right)_{eq}\ V_{TES}\ \left(T_H - T_0\right)} \end{aligned} \tag{4}$$

where t is the operating time, V the storage volume, $\left(\rho c_p\right)_{eq}$ is the storage equivalent capacitance considering the average volume contribution between the fluid and solid phase, the subscripts s and f refer to the solid and fluid phases, and T_0 is the reference temperature. One limitation of the SOC is that internal energy is not equal to availability; hence, during an idle process, the system could still have a high SOC, but the conduction could spread the temperature distribution and lower the temperatures at the top. However, according to Ref. [17], the charged energy level in the storage tank can be separated into two stages: the accumulation and the filling phases. The accumulation phase comprises the period at which SOC ≤ 0.7, when the minimum outflow losses occur. After that limit, the filling stage starts and the charging pace is reduced since the outflow temperatures increase. Using the theory of time constant, Ref. [17] defined $\tau_{0.7}$ as the time it takes the system to reach the filling stage. Considering that limit value, it is possible to state an end-of-charge criterion for the PBTES operation. This criterion can be predicted through operating and design conditions according to the methodology proposed in Ref. [17].

Entropy generation number. By following the approach introduced by Ref. [4], storage systems are components designed to store exergy; therefore, the study of their transient operation should focus on minimizing the exergy losses during the storage process. According to Ref. [4], the heating process is accompanied by internal and external irreversibilities. For the PBTES system, the internal irreversibilities correspond to the volumetric entropy generation and the exergy losses due to the finite difference between the insulation and the environment [5,14]. The external irreversibility comes from the exhaust HTF outside the PBTES and is enhanced by the early thermocline discharge due to the system's extended operation [4,14]. As proposed by Ref. [4], the entropy generation number can be defined as the ratio of the destroyed exergy and the total exergy content of the HTF that enters the storage:

$$N_s(t) = \frac{T_0 S_{gen} + X_{loss,\ \Delta T} + X_{loss,\ outflow}}{X_{max}} \tag{5}$$

Each term is defined as

$$S_{gen} = \int_{t_0}^{t} \int_V \left(k_f^* \left(\frac{\nabla T_f}{T_f}\right)^2 + k_s^* \left(\frac{\nabla T_s}{T_s}\right)^2 \right) dV\ dt \tag{6}$$

$$X_{loss,\ \Delta T} = \int_{t_0}^{t} \int_V \left(1 - \frac{T_0}{T_f}\right)\ \delta \dot{q}_{f0}'''\ dt \tag{7}$$

$$X_{loss,\ outflow} = \int_{t_0}^{t} \dot{m}\ \left(\Delta h_{outflow} - T_0 \Delta s_{outflow}\right)\ dt \tag{8}$$

$$X_{max} = \int_{t_0}^{t} \dot{m}\ \left(h_H - h_L - T_0\left(s_H - s_L\right)\right)\ dt \tag{9}$$

where k^* is the effective thermal conductivity of the respective phase, $\delta \dot{q}_{f0}'''$ is the differential heat transfer rate per unit volume between the fluid inside the tank and the environment, $\dot{m}$ is the fluid mass flow rate, $\Delta h_{outflow}$ and $\Delta s_{outflow}$ is the enthalpy and entropy difference between the outlet and the downstream or upstream temperature, depending on whether it refers to the charge or discharge process.

The entropy generation number can define criteria to determine when to charge or discharge based on the evolution of exergetic loss. For instance, the entropy generation is high at the start of the charging process since thermal gradients in the storage medium are large, but they decrease as the medium temperature rises. Consequently, the outflow losses increase due to the early thermocline extraction. Due to the behavior described, it can be assumed that a minimum N_s can define when to end the charging period. This occurs in the case study analyzed by Ref. [4], corresponding to a large liquid bath that stores the heat from a hot gas stream.

In contrast with the case studied by Ref. [4], the volumetric entropy generation in PBTES is low compared to thermal losses and external irreversibilities [14]. This is due to the low grade of mixing during the system's operation, which is caused by the different flow directions during the charge and discharge processes. In that context, the minimum entropy generation number can be obtained in small charge/discharge times, which is impractical for thermal energy storage applications.

2.2. Thermal state of PBTES systems

Considering the metrics' limitations described in the former subsection, this work proposes a novel method to monitor the thermal state of packed-bed systems. This approach considers a maximum storage potential when the hot and cold parts of the PBTES are perfectly separated. As a result, during the charging process, the hot HTF enters at T_H and exits the system at T_L for the whole period, i.e., from t_0 to t. The accumulated potential can be calculated considering the area between T_H and T_L in the time interval: $\left(T_H - T_L\right)\ \left(t - t_0\right)$. In contrast, the actual PBTES's stored-potential-in-time obeys the system's dynamics and the thermal capacity of the storage medium, so it is calculated through the enclosed area between the top and bottom temperatures: $\int_{t_0}^{t} \left(T_T\left(t'\right) - T_B\left(t'\right)\right) dt'$. Therefore, one may define an indicator based on the ratio between the actual storage capacity and its ideal, i.e.,

$$\mathrm{STP}(t) = \begin{cases} \frac{T_T(t_0) - T_B(t_0)}{T_H - T_L} & t = t_0 \\ \frac{1}{\left(T_H - T_L\right)\ \left(t - t_0\right)} \int_{t_0}^{t} \left(T_T\left(t'\right) - T_B\left(t'\right)\right) dt' & t > t_0 \end{cases} \tag{10}$$

where T_T and T_B refer to the top and bottom temperatures of the solid medium. This proposed indicator, named State of Thermal Potential (STP), can be directly measured by placing thermocouples at the top and bottom of the porous medium. This metric quantifies how near the system is from its ideal operation state. The initial condition is obtained by applying L'Hôpital's rule in the general expression. The initial condition is similar to the definition of the heat efficiency captured by solids, presented in the theory of packed-bed regenerators [18]. Although this initial stage indicates the amount of thermal energy available in the solids at time t_0, it does not depend on the STP of the previous charge/discharge process. The assumptions for the developed expression are summarized as follows:

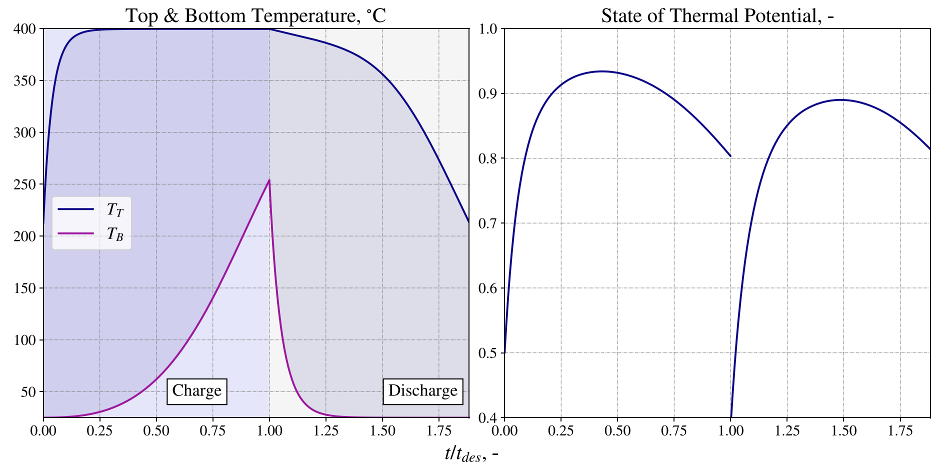

Fig. 3. Transient response of the proposed monitoring indicator. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

- The system stores sensible heat in a vertically-oriented cylindrical tank, filled with a solid medium in a packed-bed arrangement.
- The PBTES system operates between a high-temperature heat source at T_H, and a low-temperature heat sink at T_L.
- The charge and discharge process occurs in different direction; therefore, a thermocline is developed along the tank separating the hot and cold part of the storage.
- The maximum stored potential at time t is achieved when the heat transfer fluid enters to the storage at T_H and exits at T_L.

Fig. 3 reveals the dynamic behavior of the STP contrasted with the time response of the PBTES's top and bottom temperatures (T_T and T_B). The operation corresponds to a charge–discharge process with defined operating times. The colored rectangles in the plot on the left represent the ideal stored/released potential during the charge and discharge processes, whereas the actual potential is enclosed between the curves of T_T and T_B and represented by the shaded area. Consequently, the STP plotted on the right shows the transient evolution of the ratio between the two areas. For the sake of simplicity, its behavior will be explained in depth for the charging process. The curve depicted by the STP presents an increase until reaching a maximum value; then, it decreases towards the period's ending. The part before the maximum is the initial heating, in which the top solids are heated to an asymptotic value closer to the HTF inlet's temperature. Due to the temperature spread along the tank, the bottom temperature increase is delayed and is conductive predominant. The maximum value occurs when the top has reached the asymptotic temperature, and the convective heat transfer at the bottom becomes dominant against conduction. This initiates the decrease in the enclosed area as the thermocline starts to be released out of the PBTES. The discharge process follows the same principle, but it is oriented towards the solid's cooling dynamics.

2.3. Comparison between metrics

A comparison between the different approaches to monitoring the charge/discharge state of the PBTES is conducted using a case study. The design conditions for the configuration described above are listed in Table 1. The PBTES operation is determined using a previously-validated model from our research group, which was published in [14]. The system is subjected to steady boundary conditions and several operating cycles; thus, the PBTES's sizing is determined through the design conditions and a simple energy balance [8]. On the other hand, different stopping criteria are assessed for the operating conditions based on the monitoring indicators reviewed in the previous subsection. These criteria are only applied to the charging process, while the discharge process will be extended until a limit temperature is reached.

Table 2 shows the five criteria compared in the PBTES's dynamic operation, which extends the options researched by Ref. [11] by the inclusion of stopping criteria based on enhancing the PBTES's performance as a component (as Cases 3 to 5).

The PBTES's dynamics is presented in Fig. 4 for the five cases described in Table 2. The plots on the left illustrate the outflow temperature during the charge and discharge processes. The curve discontinuity is due to the end of the charging period. The flow is reversed instantly, and the outflow temperature monitoring is performed at the PBTES's top. The x-axis is the dimensionless operating time, which is the actual time divided by the design-storage hours (t_{des}). The plots on the right in Fig. 4 depict the transient behavior of the state of charge (SOC), the thermocline thickness (δ), and the state of thermal potential (STP). The entropy generation number Ns is not included due to the non-practical operating times that resulted from imposing the stopping criteria $dN_s/dt = 0$. Case 1 is used as a benchmark for the component's operation without imposing any stop criteria. The SOC behaves as a monotonic function, increasing in charge and decreasing in the discharge as the solid medium rises and releases its internal energy, respectively. The SOC reaches values near one during the charging process when the operating time equals the design-storage time, which is also coherent with the elevated outflow temperature caused by the thermocline release from the PBTES. The latter can be confirmed with the thermocline thickness dynamics. This increases in time to a maximum value, indicating that the low-temperature threshold has reached the bottom of the storage. Likewise, for the discharging process, the thermocline thickness increases due to the bottom solids cooling down. The top temperature decreases as the discharge progresses; therefore, the thermocline extends until the top threshold is reached. From that moment, the temperature decreases at an accelerated rate up to the discharging limit. Finally, the STP behaves similarly to the thermocline thickness; therefore, both indicators provide the same information through different formulation approaches.

When comparing the different stopping criteria for the charging process based on its dynamics, having no constraints in the system's operation extends the time in which the PBTES delivers heat between the process requirements, having a discharging time near its design. Conversely, imposing a cut-off temperature limits the amount of thermocline released outside the PBTES and reduces its overall operating time considerably. The same behavior is observed when setting the charging process until the thermocline thickness or the STP reaches its maximum value: even with a small value on the bottom PBTES temperature, the discharge process occurs for a small operating time, as the temperature decays with a steeper slope than a non-restricted operation (see Fig. 2). Since the charging time is controlled to retain the thermocline within the PBTES, the high-temperature zone is not fully

Table 1
Design conditions for the case study.

Parameter	Symbol	Value
Source temperature	T_H	400 °C
Sink temperature	T_L	25 °C
Thermal power demand	$\dot{Q}_{load}$	250 kW
Design storage hours	t_{des}	12 h
Design efficiency	η_0	0.8
Filler material	Density, ρ_s	3700 kg m^{-3}
	Specific heat capacity, $c_{p,s}$	952 J kg^{-1} K^{-1}
	Thermal conductivity, k_s	2.1 W m^{-1} K^{-1}
	Particle diameter, d_p	2.5 cm
Container design	Shape & Topology	Cylindrical tank with axial flow
	Aspect ratio, AR	1.2
	Material	Carbon steel

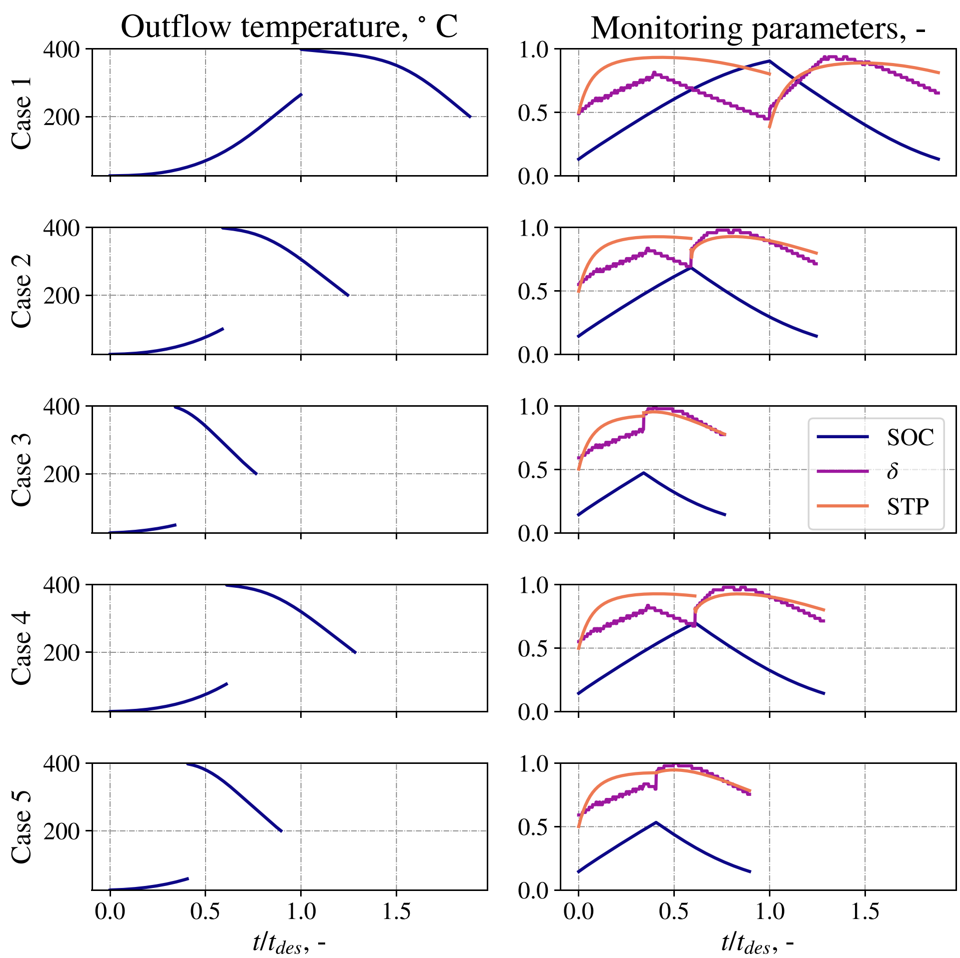

Fig. 4. PBTES dynamic operation.

Table 2
Charge stopping criteria.

Charge stopping criteria	Case tag	Mathematical definition
Reaching design charging time	1	$t = t_{des}$
Cut-off temperature	2	$T_B = T_{co,\ ch}$
Maximum thermocline length	3	$\frac{d\ \delta}{dt} = 0$
Limit the SOC to minimize outflow losses	4	SOC = 0.7
Maximum stored thermal capacity	5	$\frac{d\ \mathrm{STP}}{dt} = 0$

developed along the tank. Constraining the SOC can also be an option to limit the thermocline release out of the PBTES. Indeed, in this simple case study, the SOC limit remains fixed to ensure an operation below the filling stage. However, by using the methodology from Ref. [17], the system can be controlled using the time constant $\tau_{0.7}$ to exploit the PBTES's accumulation stage. The impacts on the operational constraints can be quantified through performance metrics, such as round-trip efficiency η_{rt}, which calculates the ratio between the energy discharged and charged. The other metric assessed is the utilization factor ζ, which quantifies the amount of useful energy released during the discharge process over its design value. Comparing these metrics, operational constraints for retaining the thermocline within the PBTES (Cases 3 and 5) improve the round-trip efficiency, but the resulting operation is far from the PBTES's design conditions. Limiting the SOC to operate the system until the end of the accumulation stage corresponds to an in-between alternative to decrease the outflow losses and operate closer to the design conditions. When using a charging cut-off temperature of 100 °C, the performance parameters are similar to Case 4. However,

Table 3
Performance of each operational constraint.

Case	η_{rt}	ζ
1	0.913	0.773
2	0.940	0.540
3	0.959	0.329
4	0.939	0.556
5	0.954	0.387

Table 4
Design data ranges of variation for sizing the PBTES system.

Design parameter	Range for analysis
T_H	400 °C to 700 °C
T_L	25 °C to 300 °C
$\dot{Q}_{load}$	100 kW to 10 000 kW
t_{des}	1 h to 16 h
η_0	0.8
AR	1.2
Filler material	Fixed: Copper slags thermophysical properties [19]
Heat transfer fluid	Fixed: Air

these performances can vary depending on the defined offset value (see Table 3).

What stands out in these comparisons is the similar operational limits and performances when using Cases 3 and 5 as stopping criteria. Thus, these indicators could be considered equivalent. Still, there are some advantages in the STP's implementation: (I) it does not need the predefined temperature threshold to constrain the thermocline limits, and (II) measurements can be performed with thermocouples located at the top and bottom of the porous medium.

3. Parametric analysis

The system's operation up to the maximum STP indicates the stopping time in which the thermocline is not released from the storage and the minimum allowed threshold temperature. This limit will depend on different design aspects that define the PBTES's size. Through a parametric analysis, it is possible to study the behavior of the STP_{max} as a function of the design conditions listed in Table 4.

The temperature profile development within the PBTES depends on the conduction, radiation, and convective heat transfer phenomena. However, to save computational costs in studying how STP_{max} correlates with design variables, conduction and radiation are neglected compared to convection since it is the main source heating and cooling the solids [20]. As a result, with all the design variables, two main parameters are computed:

$$\tau = \frac{(1-\varepsilon)\ \left(\rho c_p\right)_s\ V_{TES}}{\dot{m}c_{p,f}} \tag{11}$$

$$\mathrm{NTU} = \frac{h_v\ V_{TES}}{\dot{m}c_{p,f}} \tag{12}$$

where τ is the total time to heat the entire solid medium within the PBTES, and NTU is the number of heat transfer units. ε, ρ_s, $c_{p,s}$, and $c_{p,f}$ comes from the given design data, whereas V_{TES}, $\dot{m}$, and h_v are calculated from the provided information (see Appendix A).

The diagram shown in Fig. 5 describes the workflow to develop the parametric analysis of the PBTES operation. The different design data combinations are defined using the ranges indicated in Table 4. The model to simulate the charging process and computing the STP(t) is a simplified version of Schumann's model to reduce computational costs [21]. The details of its resolution are included in Appendix B. This analysis aims to determine the correlation between design parameters for PBTES systems and the operational conditions that define the end of the charging process. Hence, the main outcome variables are the charge cut-off temperature $\Delta T_{co,\ ch}$ and the charging time t_{ch}. These are determined from the constraint that the system operates until the STP(t) achieves its maximum value. Thus, $\mathrm{STP}\left(t_{ch}\right) = \mathrm{STP}_{max}$ and $\Delta T_{co,\ ch} = f\left(\mathrm{STP}_{max}\right)$.

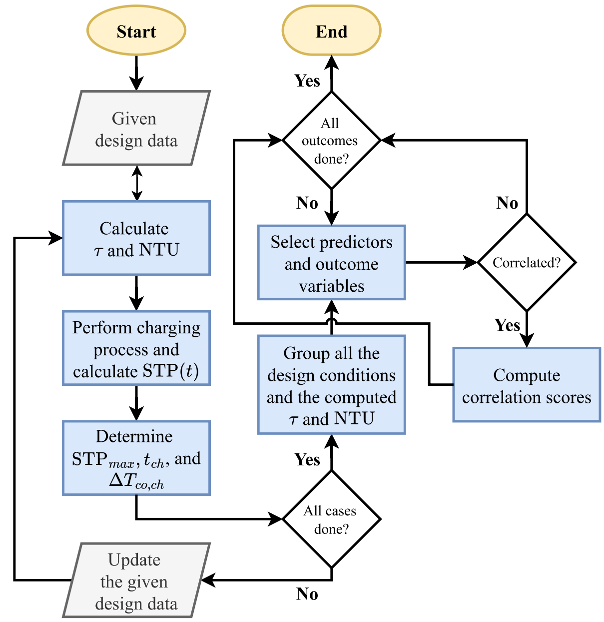

Fig. 5. Flowchart for parametric and correlation analysis.

Fig. 6 depicts the result of the STP_{max} as a function of the thermal load and the design-storage time (left and center plots). Each point in the scatter is colored according to $\Delta T_{max} = T_H - T_L$. For both $\dot{Q}_{load}$ and t_{des}, when one is constant, there is a variation on STP_{max}. This is due to the strong dependency of the other variables, such as the temperature limits. All this variability can be summarized in the NTU as a metric for heat transfer performance within the TES. After combining $\dot{Q}_{load}$, t_{des}, T_H, and T_L to calculate the NTU, it is demonstrated that all the STP_{max} values fit on a single line. This consists of a power-law correlation $\mathrm{STP}_{max} = a\mathrm{NTU}^b + c$, with a, b, and c as parameters to be estimated. Considering this proposed relationship, having a larger NTU enhances the system's proximity to its ideal operation. As these results are intuitive, the combination of the input data influences the calculated NTU. For instance, with a fixed thermal load and small ΔT_{max}, the required mass flow rate increases compared to a system that operates on a broader range of temperatures, enhancing the convective heat transfer coefficient. If the design storage time is also fixed, larger storage volumes are required to store heat when ΔT_{max} is reduced. This enlarges the solid's surface to exchange heat, and the heat transfer is improved by combining the convection effect. The charge cut-off temperature can be obtained directly from the STP_{max} by using the proposed definition. The mathematical steps are as follows. First, one uses the derivative of STP in time to relate the top and bottom temperatures with the STP:

$$\frac{d\mathrm{STP}\,(t)}{dt} = \frac{T_T\,(t) - T_B\,(t)}{\left(T_H - T_L\right)\left(t - t_0\right)} - \frac{\mathrm{STP}\,(t)}{\left(t - t_0\right)} \tag{13}$$

Evaluating $t = t_{ch}$ for the maximum STP, it can be assumed that $d\mathrm{STP}\left(t_{ch}\right)/dt = 0$, $T_T\left(t_{ch}\right) \sim T_H$, and $\mathrm{STP}\left(t_{ch}\right) = \mathrm{STP}_{max} = a\mathrm{NTU}^b + c$. Therefore, the former expression turns into the following:

$$1 - \frac{T_B\left(t_{ch}\right) - T_L}{T_H - T_L} - \left(a\mathrm{NTU}^b + c\right) = 0 \tag{14}$$

Defining $\Delta T_{co,\ ch} = T_B\left(t_{ch}\right) - T_L$ and $\Delta T_{max} = T_H - T_L$, the correlation for the cut-off temperature holds:

$$\Delta T_{co,\ ch} = \Delta T_{max}\left(1 - \left(a\mathrm{NTU}^b + c\right)\right) \tag{15}$$

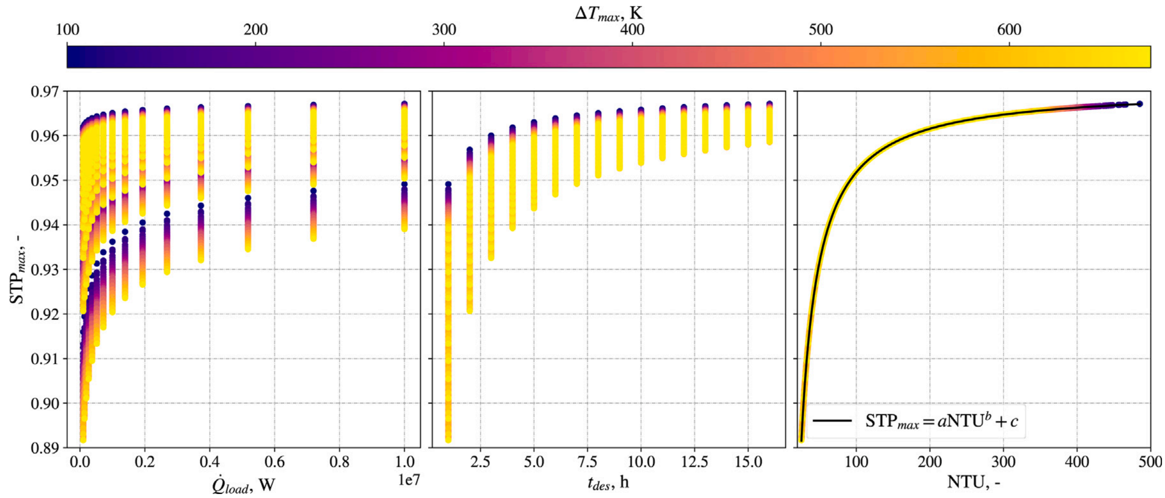

Fig. 6. Variation of maximum STP with the input design data. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

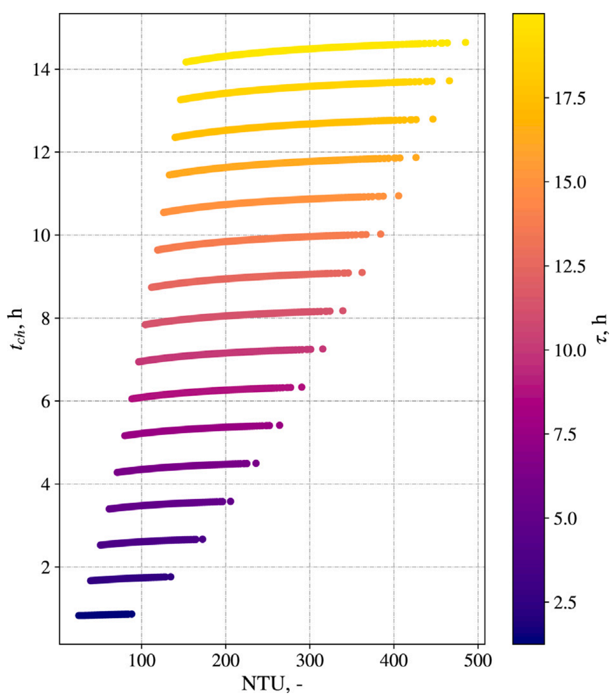

Fig. 7. Time for the maximum state of thermal potential as a function of NTU and τ.

For the charging time (t_{ch}), its variation according to τ and NTU are plotted in Fig. 7. Different contour lines represent each τ value calculated from the design data. In these contours, the charging time increases with the NTU value due to the enhanced heat transferred to the solid medium. The correlation for the charging time considers a power-law behavior with NTU and a linear tendency with τ. Intuitively, such a linear tendency should be reasonable since τ is the time that it takes to heat all the solid medium, whereas t_{ch} is the maximum time the solids can contain the thermal energy within the storage system. Therefore, from the data plotted, the proposed correlation is

$$t_{ch} = \alpha\tau\mathrm{NTU}^{\beta} \tag{16}$$

where α and β are parameters to be determined for the correlation. To obtain the parameters for Eqs. (15) and (16) the Non-linear Least Squares (NLS) method is used. Thus, the proposed correlations are fitted with the values of $\Delta T_{co,\ ch}$ and t_{ch}, which can be obtained through

Table 5
Correlations for the operational parameters.

Outcome	Correlation	Parameters	RMSE
$\Delta T_{co,\ ch}$	$\Delta T_{max}\left(1-\left(a\mathrm{NTU}^{b}+c\right)\right)$	$a=-2.3189$ $b=-1.0436$ $c=0.9707$	6.4034×10^{-3} K
t_{ch}	$\alpha\tau\ \mathrm{NTU}^{\beta}$	$\alpha=0.5949$ $\beta=0.0347$	45.8832 s

numerical simulation when constraining the charging process until $d\mathrm{STP}/dt = 0$. The parameter estimation results are presented in Table 5. The coefficients of these correlations are valid for NTU $\in$ [25.44, 485.13], $\Delta T_{max} \in$ [100, 675] K, and $\tau \in$ [4499, 71993] s. Both correlations show small RMSE values; therefore, predicting these values given the inputs should be precise for assessing the PBTES's operational limits.

Fig. 8 plots the operational limits obtained from the simulation versus the ones calculated from the correlations. Furthermore, the identity line is included to assess the proximity of the points in the graph, and each of them is colored with the residual values computed as the difference between the predicted and the simulated values, i.e., $res = pred - sim$. As expected, most points fit with the identity due to the low residual values. When compared with the numerical results, these correlations are precise in the given range of analysis.

4. Potential applications

This section describes how this proposed monitoring indicator can be employed to describe and obtain the operating characteristics of a PBTES system, which can be used for simulation, experimental studies, or the implementation of controlled PBTES systems. The first two subsections explain how this mathematical approach can be implemented for control, simulation, and optimization purposes. The third section relates this study to the engineering context as a summary of contributions.

4.1. Constraining operation

As discussed in Section 2.3, the development of the charging process until the STP reaches its maximum value retains the thermocline within the storage and enhances the round-trip efficiency. However, the system's discharge period becomes shorter than that of the unconstrained operating mode, which is expected since not all solids are completely

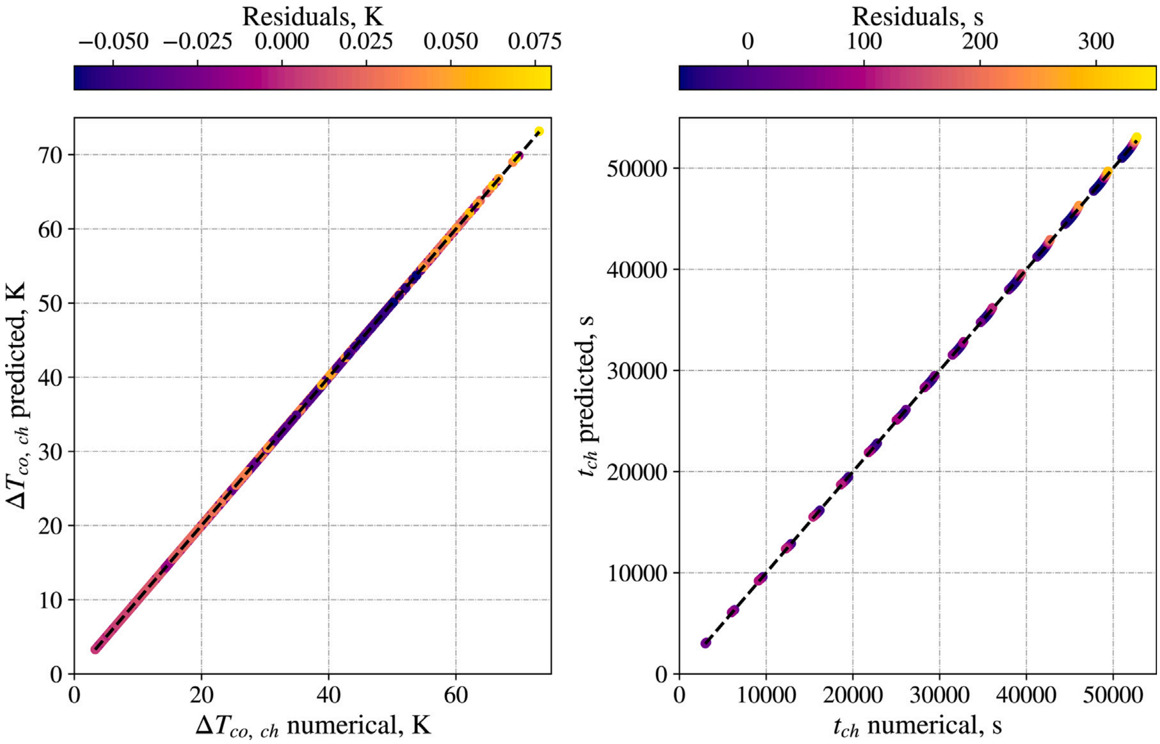

Fig. 8. Comparison between simulated and predicted values of $\Delta T_{co,\,ch}$ and t_{ch}. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

heated in the PBTES. During the simulation of a PBTES system or the operation of an existing installation, the top and bottom temperatures of the porous matrix should be measured to calculate the STP evolution in time. Then, from the numerical calculations or the data acquisition system, the derivative $d\mathrm{STP}(t)/dt$ can be calculated and operate the system until it equals zero. This can be imposed as a restriction of the numerical method or a condition in the Programmable Logic Controller (PLC) employed.

Another simple method that can be implemented for monitoring and imposing the stopping condition is considering the mathematical definition of $d\mathrm{STP}(t)/dt$ (see Eq. (13)). At STP_{max}, this value is equal to the system's effectiveness when $t = t_{ch}$:

$$\mathrm{STP}\left(t_{ch}\right) = \frac{T_T\left(t_{ch}\right) - T_B\left(t_{ch}\right)}{T_H - T_L} = \mathrm{STP}_{max} \tag{17}$$

Therefore, if the $\mathrm{STP}(t)$ and $\left(T_T(t) - T_B(t)\right)/\left(T_H - T_L\right)$ are monitored over time, the charge can be programmed to end when these functions are equal. Considering the design parameters listed in Table 1, Fig. 9 presents the PBTES's operation during a charge cycle under two sets of boundary conditions (BC) with a steady (left) and a dynamic (right) inlet mass flow rate. Each plot shows the dynamic behavior of the STP and the ratio $\left(T_T - T_B\right)/\left(T_H - T_L\right)$. In both cases, the stopping criterion $d\mathrm{STP}(t)/dt = 0$ is implemented, and the depicted curves demonstrate that the charge stops when the system's effectiveness is equal to the STP. The simulation with dynamic boundary conditions considers a proportional variation of the design mass flow rate in one-hour intervals. It should be noted that the system's effectiveness differs from the STP, as the former is an instantaneous ratio between the measured temperatures at the top and bottom and the maximum operating temperature range; therefore, the maximum value of effectiveness will be reached when the PBTES has the largest temperature difference, occurring when the top solids are close to the source temperature. Conversely, the STP is an enclosed-area ratio and is maximized when the top and bottom temperature dynamics are near to the ideal. This definition considers the physics within the PBTES, as the maximum occurs when the bottom part of the storage has a dominant convective heat transfer in its temperature increase.

4.2. Definition of operational parameters

With the correlations obtained through the methodology presented in Section 3, the operational parameters $\Delta T_{co,\,ch}$ and t_{ch} can be predicted from design data. This can be used for the following:

- Obtaining information to optimize the system from design and operational perspectives instead of having the cut-off temperature as unknown. This makes it possible to reduce the dimensions of the variables in an optimization algorithm.
- Tuning parameters to control the charging process if the user aims to maximize the PBTES's efficiency.
- Having an estimate of the required time that it takes for the thermocline to be released from the tank and assess if this operational approach matches the design requirements. If not, the $\tau_{0.7}$ value should be evaluated to operate the system until the filling stage ends [17] or another operational strategy should be implemented.

Between consecutive cycles, the development of the temperature profile will remain similar to the 5th cycle [14]; therefore, the cut-off temperature that maximizes the STP will be different from the one predicted with the design conditions. This is due to the initial condition after each operating cycle and the thermal losses affecting the system's operation. Table 6 lists the NTU and τ values calculated from input data in Table 1, as well as the results for predicting the $\Delta T_{co,\,ch}$ and t_{ch}. These operational parameters were implemented to run the simulation on seven charge–discharge cycles, also considering the input design data in Table 1. Here, two operational constraints were assessed: (I) the predicted $\Delta T_{co,\,ch}$ as a charging cut-off condition and (II) the other considering the constraint $d\mathrm{STP}/dt = 0$. For both cases, the discharge extends until the system's outlet temperature decreases to 200 °C.

Fig. 10 compares the two aforementioned operational approaches to constrain the system's charge–discharge. The plot on the left corresponds to the STP's evolution over time. Both curves start at zero in the normalized time scale to highlight the different times in which the charging process ends. Considering the stopping condition of $d\mathrm{STP}/dt =$

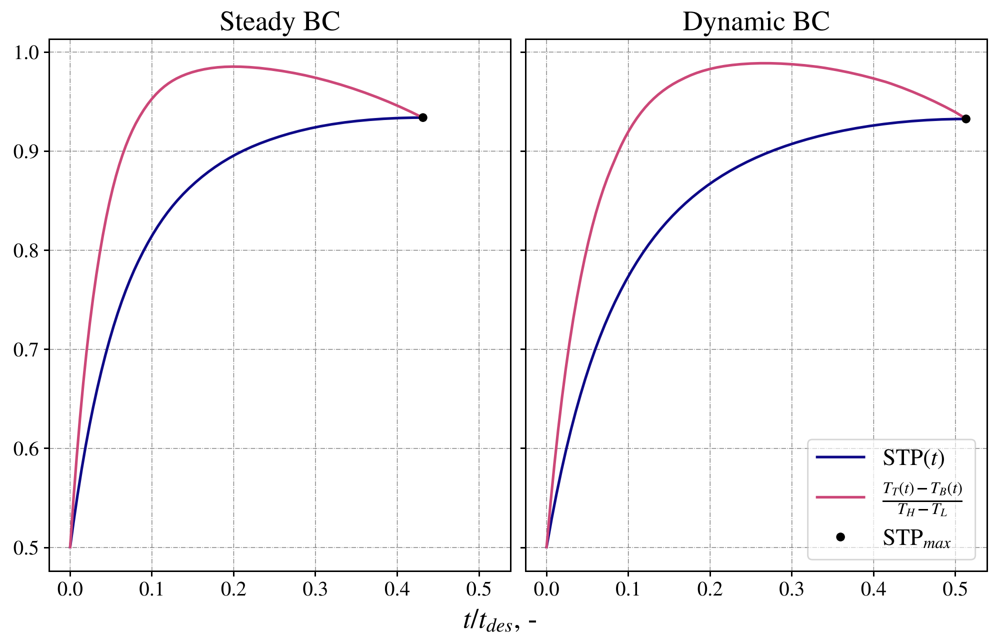

Fig. 9. Monitoring of the PBTES operation.

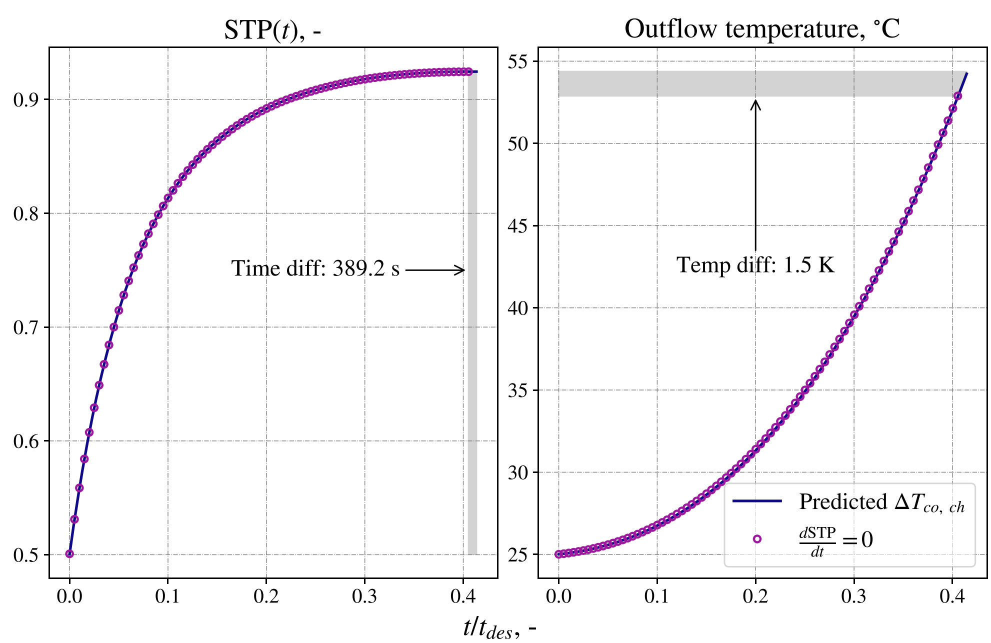

Fig. 10. Thermal state and outflow temperature during the charging process after 7 charge–discharge cycles.

Table 6
Predicted operating parameters from case study design conditions listed in Table 1.

NTU	τ	$\Delta T_{co,\ ch}$	t_{ch}
40.2709	41 242.1041 s	29.3675 K	27 891.8919 s

0, the charge ends 389 s earlier than using the predicted cut-off temperature, which will provide an offset on the subsequent cycles. The differences are also small when analyzing the outflow temperature to compare the two approaches, with a 1.5 K of temperature offset. Considering the magnitude of the temperature and time differences resulting at the end of the charging process, the predicted cut-off temperature and the $d\mathrm{STP}/dt = 0$ are equivalent criteria for ending the system's charge.

4.3. Summary

This summary encapsulates the key contributions of the research presented in this article. Through the mathematical approach to define the thermal state of packed-bed thermal energy storage systems, several outcomes have emerged that can be applied in the engineering development of these components. Table 7 provides specific contributions to the optimization, control, and implementation of PBTES systems. With the information obtained from the correlations proposed in this research and the proposal of this monitoring function, the user can manage the system's operation closer to its theoretical storage capacity without requiring high-resolution temperature profiles or complex instrumentation. The three topics listed in Table 7 support the engineering development and deployment of PBTES technologies, both in simulation and practical implementation.

Table 7
Contributions to thermal energy storage systems' engineering.

Application	Contribution
Optimization	Reduction of optimization variables, as $\Delta T_{co,\ ch}$ can be obtained from Eq. (15) which is usually an unknown in optimization framework [13,14]
Control	(1) Programmable stopping criteria from the dynamic function STP(t), avoiding early extraction of the thermocline. (2) Using the empirical correlations for $\Delta T_{co,\ ch}$ or t_{ch} to define control set points without real-time computation.
Implementation	Guideline for monitoring, setting the use of only two thermocouples (located at the top and bottom part of the porous medium) are required to monitor the state of the PBTES unit

Finally, the study's limitations mainly relate to using the empirical correlations derived from the State of Thermal Potential sensitivity analyses. First, Eqs. (15) and (16) are only valid for NTU $\in$ [25.44, 485.13], $\Delta T_{max} \in$ [100, 675] K, and $\tau \in$ [4499, 71993] s. Furthermore, these were obtained from an "empty" state PBTES system that evolves from a steady boundary condition (i.e., constant mass-flow rate). However, these predicted limits can serve as initial control set points. Still, this study develops a method for monitoring the system's state with a stopping condition for the charging process without explicitly knowing the cut-off limits. Its use will depend on the available instrumentation and the user's or system operator's operational philosophy.

5. Conclusions

The present study presents a comprehensive analysis of a packed-bed thermal energy storage system's dynamic operation. It reviews various metrics commonly used to monitor the system's thermal state and, through a mathematical model, proposes a novel approach to characterize the stored energy content. Such an approach compares the actual stored potential and the storage in ideal operating conditions. The proposed metric, called "State of Thermal Potential" (STP(t)), quantifies the ratio between the actual thermal energy stored (calculated from real-time temperature measurements at the top and bottom of the bed) and the theoretical maximum storage under ideal thermocline conditions.

The evolution of STP(t) over time reveals a well-defined maximum during the charging process, corresponding to the moment when the thermocline begins to be released from the system. By using the condition $d\mathrm{STP}/dt = 0$ as the stopping criterion for the charging process, the system ensures high thermal efficiency since the solid medium accumulates most of the thermal energy injected into the system. This method provides a simple yet effective operational tool, requiring only two temperature measurements to build the monitoring function. Therefore, it can be easily implemented in a data acquisition system or a programmable logic controller without the need for complex instrumentation.

The proposed operational limit also defines the time and cut-off temperature constraints (t_{ch} and $\Delta T_{co,\ ch}$), which are commonly unknown in packed-bed operation analyses and often left to empirical adjustment. Through a parametric study of the charging process, different design conditions were assessed, and it was possible to correlate these operational parameters with the design inputs, condensed into two main predictors: the NTU and the packed bed's characteristic time τ. With these correlations, it is possible to:

- Predict control set points based on design data without requiring simulation or calibration.
- Simplify optimization problems, as the temperature constraints can be directly estimated from design parameters.

This work provides a new approach to measuring the thermal state of a packed-bed thermal energy storage system, aiming to exploit its potential and guarantee an efficient operation. It complements the available information on the operational characteristics of these systems and offers a new perspective for measuring, quantifying, and constraining the packed-bed operation. Further research should focus on implementing this metric into an integrated system to look into the control methods, dispatch optimization, and design implications.

CRediT authorship contribution statement

Ignacio Calderón-Vásquez: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Felipe G. Battisti:** Writing – review & editing, Methodology, Investigation. **Rodrigo Escobar:** Writing – review & editing, Supervision, Project administration. **José M. Cardemil:** Writing – review & editing, Supervision, Project administration.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Grammarly AI Writing Assistance in order to check the article's grammar. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Calculation of τ and NTU from input design data

considering the input design data listed at Table 4 the calculation of the main parameters τ and NTU defined in Eqs. (11) and (12), requires computing the operating mass flow rate $\dot{m}$, the storage volume V_{tes}, and the volumetric heat transfer coefficient h_v. From the design input data, we can calculate the HTF properties using the thermal reservoir's temperature value; thus, ρ_f, $c_{p,f}$, k_f, and μ_f are assumed as known. On the other hand, the fluid's mass flow rate can be calculated by satisfying the thermal load's power requirements:

$$\dot{m} = \frac{\dot{Q}_{load}}{h_H - h_L} \tag{A.1}$$

where h_H and h_L is the fluid's enthalpy calculated from the high- and low-temperature thermal reservoirs. The storage volume is calculated from the input design data as

$$V_{TES} = \frac{\dot{Q}_{load}\ t_{des}}{\eta_0 \left(\rho c_p\right)_{eq} \left(T_H - T_L\right)} \tag{A.2}$$

where $\left(\rho c_p\right)_{eq} = \varepsilon \left(\rho c_p\right)_f + (1-\varepsilon)\left(\rho c_p\right)_s$ is the equivalent capacity between the fluid's and solid phase with ε the packed-bed's void fraction. Finally, for the volumetric heat transfer coefficient, the correlation defined by Ref. [22] for the fluid-to-particle Nusselt number Nu is employed; therefore, its calculation proceeds as follows:

$$A_{pb} = \left(\frac{\pi}{4}\right)^{1/3}\left(\frac{V_{TES}}{\mathrm{AR}}\right)^{2/3} \tag{A.3}$$

$$G = \dot{m}/A_{pb} \tag{A.4}$$

$$\mathrm{Re} = \frac{G\ d_p}{\mu_f} \tag{A.5}$$

$$\mathrm{Pr} = \frac{c_{p,f}\,\mu_f}{k_f} \tag{A.6}$$

$$\mathrm{Nu} = 2 + 1.1\ \mathrm{Re}^{0.6}\ \mathrm{Pr}^{1/3} \tag{A.7}$$

$$h_v = \left(\frac{6k_f\,(1-\varepsilon)}{d^2}\right)\mathrm{Nu} \tag{A.8}$$

where A_{pb} is the cross section of the TES system, G the mass flux rate, Re the Reynolds number, Pr the Prandtl number, and Eq. (A.7) is the expression proposed in [22].

Appendix B. Schumann's model resolution

Schumann's model neglects the thermal conductivity for both fluid and solid phases. Also, the heat capacitance of the fluid's phase is assumed to be small compared to the solid; therefore, this two-phase heat transfer model remains as

$$\dot{m}c_{p,f}\,\frac{\partial F}{\partial x} = -h_v A_{pb}(F - S) \tag{B.1}$$

$$(1-\varepsilon)\,\rho_s c_{p,s}\,\frac{\partial S}{\partial t} = h_v(F - S) \tag{B.2}$$

$$\mathrm{NTU} = \frac{h_v A_{pb} L}{\dot{m}c_{p,f}} \tag{B.3}$$

$$\tau = \frac{(1-\varepsilon)\,\rho_s c_{p,s} A_{pb} L}{\dot{m}c_{p,f}} \tag{B.4}$$

here, we are using F and S as the fluid and solid temperature. A_{pb} is the storage cross section and L its length; hence, the storage volume results from $V_{tes} = A_{pb}L$. Replacing the definition of the NTU and the time constant τ in Eqs. (B.1) and (B.2), the PDE system yield:

$$\frac{\partial F}{\partial x} = -\frac{\mathrm{NTU}}{L}\,(F - S) \tag{B.5}$$

$$\frac{\partial S}{\partial t} = \frac{\mathrm{NTU}}{\tau}\,(F - S) \tag{B.6}$$

Using the finite difference method for the spatial derivatives: $\frac{\partial F}{\partial x} = \frac{F_k - F_{k-1}}{\Delta x}$ with $\Delta x = \frac{L}{N-1}$ and N the number of discretization nodes, we can find F_k in terms of the other variables. We define $F_0 = T_H$ for simplicity, and the following formula is for $k \geq 1$.

$$F_k - F_{k-1} = -\frac{\mathrm{NTU}}{N-1}\left(F_k - S_k\right) \tag{B.7}$$

$$F_k = \frac{1}{1 + \frac{\mathrm{NTU}}{N-1}}\left(F_{k-1} + \frac{\mathrm{NTU}}{N-1} S_k\right) \tag{B.8}$$

Replacing the kth term in the equation for S, we can obtain a differential equation for the kth derivative of S, which can be solved analytically with the integrating factor.

$$S_k(t) = \exp\left(-\frac{\mathrm{NTU}}{\tau\left(1 + \frac{\mathrm{NTU}}{N-1}\right)}t\right)\left(S_k(0) + \int_{t=0}^{t} \frac{\mathrm{NTU}}{\tau\left(1 + \frac{\mathrm{NTU}}{N-1}\right)} \exp\left(\frac{\mathrm{NTU}}{\tau\left(1 + \frac{\mathrm{NTU}}{N-1}\right)}t\right) F_{k-1}\,dt\right) \tag{B.9}$$

Thus, the STP can be calculated using the expressions from S_1 and S_N using the general formula for $S_k(t)$ presented at Eq. (B.9).

$$\mathrm{STP}(t) = \frac{1}{\left(T_H - T_L\right)\left(t - t_0\right)} \int_{t_0}^{t} \left(S_1(t') - S_N(t')\right) dt' \tag{B.10}$$

Data availability

No data was used for the research described in the article.

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